DHATHATHREYAN NATARAJAN

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EDUCATION

National University of Singapore — **Aug 2024 – May 2028**
Bachelor of Science in Data Science and Economics — *Singapore*

St. Joseph's Institution — **Feb 2020 – Nov 2021**
International Baccalaureate Diploma — *Singapore*

- IB score: **42/45** (Higher Level: Mathematics, Economics, Chemistry). Founding Vice President, Education and Career Guidance Committee; ran an Excel training workshop for children at a Chinese Development Association council.

WORK EXPERIENCE

Pacific Carriers Limited — **May 2025 – Jul 2025**
Analysis & Research Intern, Business Management Office — *Singapore*

- Designed and built a **Python optimiser** (Pandas, NumPy) for **FuelEU Maritime, IMO and EU ETS** compliance, streamlining the team's emissions and carbon-cost calculations across the fleet.
- Developed an Excel fleet database with an embedded **DCF model** to value fuel options, retrofits and energy-saving devices; traced and resolved inconsistencies in full-fleet outputs before results reached senior management.
- Researched long-term vessel retrofit strategies and the transition to alternative fuels such as biofuels, turning the analysis into recommendations for **senior stakeholders including the CFO**.

Plutus Corporate Services Pte Ltd — **Apr 2024 – Jul 2024**
Accountant — *Singapore*

- Managed monthly accounts payable, accounts receivable and **bank reconciliations** in XERO; prepared ledger balances and P&L statements in Excel.
- Processed bills, invoices and bank statements in Excel for import into XERO, maintaining accurate client records.

Singapore Armed Forces — **Dec 2022 – Nov 2023**
Combat Engineer Officer — *Singapore / Australia*

- Commanded a unit through training and **2 major evaluations**; planned and led explosives training for **100 soldiers**.
- Represented the unit in Exercise Wallaby, a 3-week overseas exercise in Australia, coordinating with other units.

PROJECTS & COMPETITIONS

Algorithmic Trading System · Personal Project — **Ongoing**

- Building a **Python** system that pulls live and historical equity data through broker and market-data APIs and generates rule-based signals (moving averages, RSI); designing parameter-sensitivity and out-of-sample tests before backtesting.

Weather Humidity Prediction · Team Member — **Mar 2025 – May 2025**

- Built **Random Forest, XGBoost, LSTM and hybrid LSTM-XGBoost** models in Python to forecast humidity at 1-hour, 6-hour and 24-hour horizons; wrote the Pandas preprocessing pipeline and evaluated inputs with stationarity tests and correlation analysis.

Maritime Information RAG Pipeline · with peers

- Built a **retrieval-augmented generation (RAG)** pipeline that returns maritime information in response to keyword queries.

NTU x MOM Essay Competition · Participant — **Jun 2025 – Jul 2025**

- Built a **machine learning similarity index** to select comparator countries, then proposed policies to raise Singapore's female labour force participation, adapted to local demographics and culture.

Temasek-NUS Stock Pitch Competition 2024 · Team Leader — **Aug 2024**

- Led a team in researching investment opportunities; built a **DCF valuation** and a football field chart benchmarking 5 competitors.

Tech for Good 2022 · Team Member — **Apr 2022 – Sep 2022**

- Developed a technology solution for the hard of hearing with a team of 5, working with a member of that community from concept to final showcase.

SKILLS

Programming	Python (Pandas, NumPy), R
Data & ML	Random Forest, XGBoost, LSTM, time-series analysis, stationarity testing, RAG pipelines
Finance	DCF valuation, comparable company analysis, financial statements, bookkeeping, XERO
Excel	Financial modelling, pivot tables and charts, formulas, data analysis
Languages	English (native), Tamil (native)